

Question:

Assume that B Corporation has the following total profit function equation: Total Profit = $\pi = -\$3,000 - \$2,400Q + \$350Q^2 - \$8.333Q^3$ a) Compute the marginal profit b) Determine its output quantities

Answer Choices:

- A. 22 and 7 units
- B. 21 and 8 units
- C. 23 and 6 units
- D. 26 and 2 units
- E. 24 and 4 units
- F. 25 and 3 units
- G. 20 and 5 units
- H. 27 and 1 unit
- I. 18 and 10 units
- J. 19 and 9 units

Reasoning + Answer:

<reasoning process here>

We are given the total profit function :

$$\pi = -3000 - 2400Q + 350Q^2 - 8.333Q^3$$

a) To compute the marginal profit , we take the derivative of the total profit function with respect to Q :

$$\text{Marginal Profit (MP)} = d\pi/dQ$$

So ,

$$MP = d/dQ [-3000 - 2400Q + 350Q^2 - 8.333Q^3]$$

Differentiate term by term :

- d/dQ (-3000) = 0
- d/dQ (-2400Q) = -2400
- d/dQ (350Q²) = 700Q
- d/dQ (-8.333Q³) = -25Q² (since 8.333 $\times$ 3 = 25)

So ,

$$\text{Marginal Profit} = -2400 + 700Q - 25Q^2$$

That's the marginal profit function .

b) To determine the output quantities , we need to find the quantity Q where marginal profit is zero (since that's where profit is maximized , assuming the firm is a profit-maximizing firm).

Set MP = 0 :

$$-2400 + 700Q - 25Q^2 = 0$$

Multiply both sides by -1 to make it easier:

$$2400 - 700Q + 25Q^2 = 0$$

Rewrite :

$$25Q^2 - 700Q + 2400 = 0$$

We can simplify this equation by dividing all terms by 25 :

$$Q^2 - 28Q + 96 = 0$$

Now solve the quadratic equation :

$$\begin{aligned} Q &= [28 \pm \sqrt{(28^2 - 4 \times 1 \times 96)}] / 2 \\ &= [28 \pm \sqrt{(784 - 384)}] / 2 \\ &= [28 \pm \sqrt{400}] / 2 \\ &= [28 \pm 20] / 2 \end{aligned}$$

So :

$$\begin{aligned} Q &= (28 + 20) / 2 = 48 / 2 = 24 \\ Q &= (28 - 20) / 2 = 8 / 2 = 4 \end{aligned}$$

Wait this gives us two values : Q = 24 and Q = 4 .

But the answer options are pairs like " 22 and 7 units ", " 21 and 8 units ", etc .

This suggests that the output quantities are two values , perhaps both are valid solutions ? But typically , for profit maximization , only one Q is optimal .

Wait but we found that marginal profit is zero at Q = 24 and Q = 4 ?

But check : the profit function is cubic , and if the marginal profit is zero at two points , then these are turning points . One may be a maximum and one a minimum .

But since the profit function has negative cubic term (-8.333Q³), profit will eventually decrease . So likely , one of these is a local maximum .

We should verify which of these gives a maximum (i.e., check second derivative):

Second derivative (d² π /dQ²) = derivative of marginal profit :

$$\begin{aligned} MP &= -2400 + 700Q - 25Q^2 \\ d^2\pi/dQ^2 &= 700 - 50Q \end{aligned}$$

At Q = 4 :

$$d^2\pi/dQ^2 = 700 - 200 = 500 > 0 \implies \text{local minimum}$$

At Q = 24 :

$$d^2\pi/dQ^2 = 700 - 50 \times 24 = 700 - 1200 = -500$$